MSG No. 125.369 $P4/nb'm'$ [Type III, tetragonal]

Table 1: Wyckoff site: `2a`, site symmetry: `42'2'`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, 0\right]$	`[1,2,3,4,9,10,11,12]`
2	$\left[\frac{3}{4}, \frac{3}{4}, 0\right]$	`[5,6,7,8,13,14,15,16]`

Table 2: Wyckoff site: `2b`, site symmetry: `42'2'`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[1,2,3,4,9,10,11,12]`
2	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[5,6,7,8,13,14,15,16]`

Table 3: Wyckoff site: `2c`, site symmetry: `-42'm'`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	`[1,4,6,7,9,10,15,16]`
2	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	`[2,3,5,8,11,12,13,14]`

Table 4: Wyckoff site: `2d`, site symmetry: `-42'm'`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[1,4,6,7,9,10,15,16]`
2	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[2,3,5,8,11,12,13,14]`

Table 5: Wyckoff site: `4e`, site symmetry: `..2'/m'`

No.	position	mapping
1	$[0, 0, 0]$	`[1,5,11,15]`
2	$\left[\frac{1}{2}, 0, 0\right]$	`[2,6,10,14]`
3	$\left[0, \frac{1}{2}, 0\right]$	`[3,7,9,13]`
4	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[4,8,12,16]`

Table 6: Wyckoff site: `4f`, site symmetry: `..2'/m'`

No.	position	mapping
1	$\left[0, 0, \frac{1}{2}\right]$	`[1,5,11,15]`
2	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[2,6,10,14]`
3	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[3,7,9,13]`
4	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[4,8,12,16]`

Table 7: Wyckoff site: `4g`, site symmetry: `4..`

No.	position	mapping
1	$\left[\frac{1}{4},\ \frac{1}{4},\ z\right]$	`[1,2,3,4]`
2	$\left[\frac{3}{4},\ \frac{3}{4},\ -z\right]$	`[5,6,7,8]`
3	$\left[\frac{1}{4},\ \frac{1}{4},\ -z\right]$	`[9,10,11,12]`
4	$\left[\frac{3}{4},\ \frac{3}{4},\ z\right]$	`[13,14,15,16]`

Table 8: Wyckoff site: `4h`, site symmetry: `2.m'm'`

No.	position	mapping
1	$\left[\frac{3}{4},\ \frac{1}{4},\ z\right]$	`[1,4,15,16]`
2	$\left[\frac{1}{4},\ \frac{3}{4},\ z\right]$	`[2,3,13,14]`
3	$\left[\frac{1}{4},\ \frac{3}{4},\ -z\right]$	`[5,8,11,12]`
4	$\left[\frac{3}{4},\ \frac{1}{4},\ -z\right]$	`[6,7,9,10]`

Table 9: Wyckoff site: `8i`, site symmetry: `..2'`

No.	position	mapping
1	$[x,\ x,\ 0]$	`[1,11]`
2	$\left[\frac{1}{2}-x,\ x,\ 0\right]$	`[2,10]`
3	$\left[x,\ \frac{1}{2}-x,\ 0\right]$	`[3,9]`
4	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ 0\right]$	`[4,12]`
5	$[-x,\ -x,\ 0]$	`[5,15]`
6	$\left[x+\frac{1}{2},\ -x,\ 0\right]$	`[6,14]`
7	$\left[-x,\ x+\frac{1}{2},\ 0\right]$	`[7,13]`
8	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ 0\right]$	`[8,16]`

Table 10: Wyckoff site: `8j`, site symmetry: `..2'`

No.	position	mapping
1	$\left[x,\ x,\ \frac{1}{2}\right]$	`[1,11]`
2	$\left[\frac{1}{2}-x,\ x,\ \frac{1}{2}\right]$	`[2,10]`
3	$\left[x,\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	`[3,9]`
4	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	`[4,12]`
5	$\left[-x,\ -x,\ \frac{1}{2}\right]$	`[5,15]`
6	$\left[x+\frac{1}{2},\ -x,\ \frac{1}{2}\right]$	`[6,14]`
7	$\left[-x,\ x+\frac{1}{2},\ \frac{1}{2}\right]$	`[7,13]`
8	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}\right]$	`[8,16]`

Table 11: Wyckoff site: `8k`, site symmetry: `.2'.`

No.	position	mapping
1	$\left[x,\ \frac{1}{4},\ 0\right]$	`[1,9]`
2	$\left[\frac{1}{4},\ x,\ 0\right]$	`[2,11]`

continued ...

Table 11

No.	position	mapping
3	$\left[\frac{1}{4},\ \frac{1}{2}-x,\ 0\right]$	[3,12]
4	$\left[\frac{1}{2}-x,\ \frac{1}{4},\ 0\right]$	[4,10]
5	$\left[-x,\ \frac{3}{4},\ 0\right]$	[5,13]
6	$\left[\frac{3}{4},\ -x,\ 0\right]$	[6,15]
7	$\left[\frac{3}{4},\ x+\frac{1}{2},\ 0\right]$	[7,16]
8	$\left[x+\frac{1}{2},\ \frac{3}{4},\ 0\right]$	[8,14]

Table 12: Wyckoff site: 8l, site symmetry: .2'.

No.	position	mapping
1	$\left[x,\ \frac{1}{4},\ \frac{1}{2}\right]$	[1,9]
2	$\left[\frac{1}{4},\ x,\ \frac{1}{2}\right]$	[2,11]
3	$\left[\frac{1}{4},\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	[3,12]
4	$\left[\frac{1}{2}-x,\ \frac{1}{4},\ \frac{1}{2}\right]$	[4,10]
5	$\left[-x,\ \frac{3}{4},\ \frac{1}{2}\right]$	[5,13]
6	$\left[\frac{3}{4},\ -x,\ \frac{1}{2}\right]$	[6,15]
7	$\left[\frac{3}{4},\ x+\frac{1}{2},\ \frac{1}{2}\right]$	[7,16]
8	$\left[x+\frac{1}{2},\ \frac{3}{4},\ \frac{1}{2}\right]$	[8,14]

Table 13: Wyckoff site: 8m, site symmetry: ..m'

No.	position	mapping
1	$[x,\ -x,\ z]$	[1,15]
2	$\left[x+\frac{1}{2},\ x,\ z\right]$	[2,14]
3	$\left[-x,\ \frac{1}{2}-x,\ z\right]$	[3,13]
4	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ z\right]$	[4,16]
5	$[-x,\ x,\ -z]$	[5,11]
6	$\left[\frac{1}{2}-x,\ -x,\ -z\right]$	[6,10]
7	$\left[x,\ x+\frac{1}{2},\ -z\right]$	[7,9]
8	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ -z\right]$	[8,12]

Table 14: Wyckoff site: 16n, site symmetry: 1

No.	position	mapping
1	$[x,\ y,\ z]$	[1]
2	$\left[\frac{1}{2}-y,\ x,\ z\right]$	[2]
3	$\left[y,\ \frac{1}{2}-x,\ z\right]$	[3]
4	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z\right]$	[4]
5	$[-x,\ -y,\ -z]$	[5]
6	$\left[y+\frac{1}{2},\ -x,\ -z\right]$	[6]
7	$\left[-y,\ x+\frac{1}{2},\ -z\right]$	[7]
8	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ -z\right]$	[8]
9	$\left[x,\ \frac{1}{2}-y,\ -z\right]$	[9]

continued ...

Table 14

No.	position	mapping
10	$\left[\frac{1}{2}-x,\, y,\, -z\right]$	[10]
11	$[y,\, x,\, -z]$	[11]
12	$\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, -z\right]$	[12]
13	$\left[-x,\, y+\frac{1}{2},\, z\right]$	[13]
14	$\left[x+\frac{1}{2},\, -y,\, z\right]$	[14]
15	$[-y,\, -x,\, z]$	[15]
16	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, z\right]$	[16]